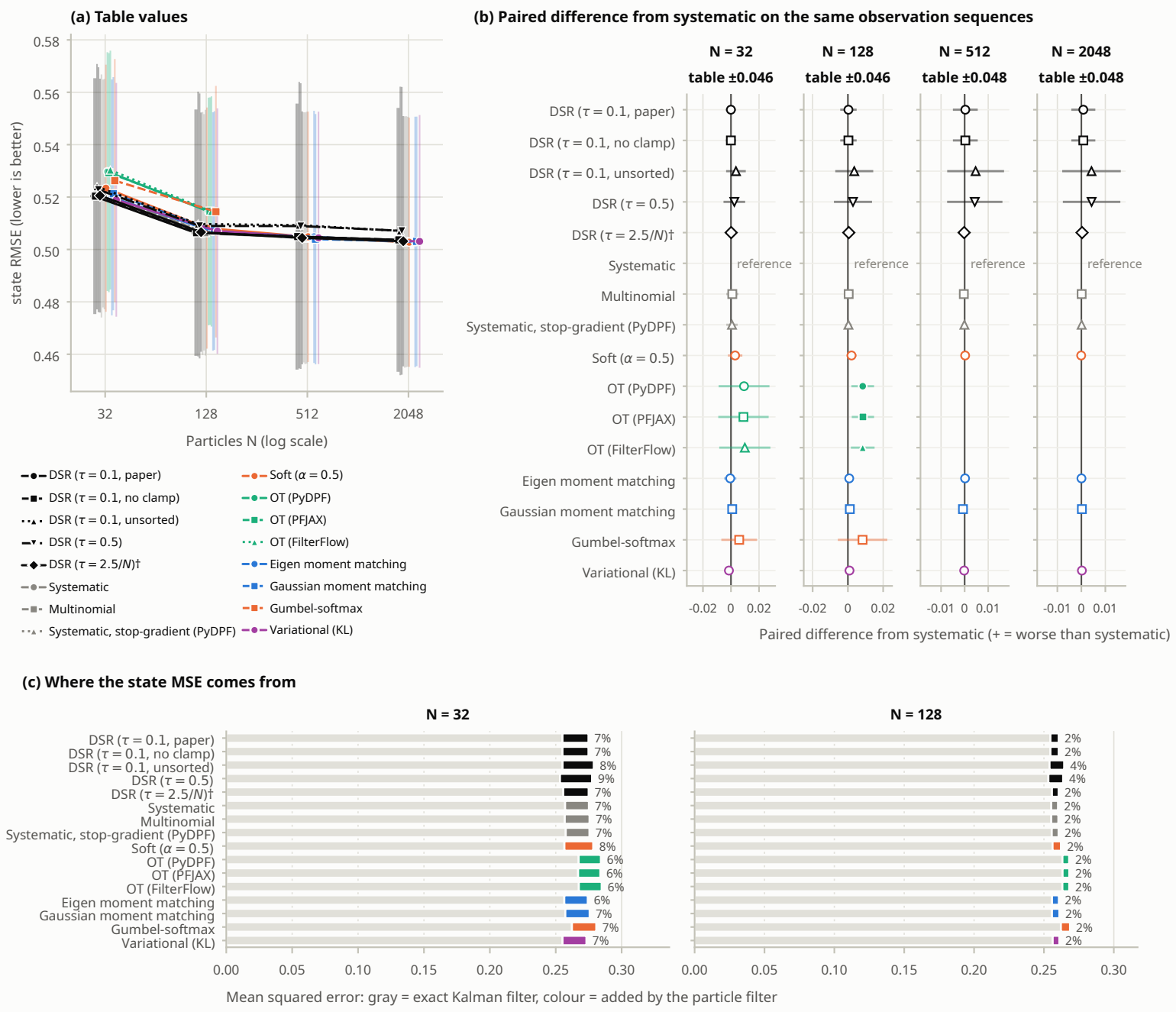

(a) Table values
state RMSE (lower is better)
0.58
0.56
0.54
0.52
0.50
0.48
0.46
32
128
512
2048
Particles N (log scale)
DSR (τ = 0.1, paper)
DSR (τ = 0.1, no clamp)
DSR (τ = 0.1, unsorted)
DSR (τ = 0.5)
DSR (τ = 2.5/N)†
Systematic
Multinomial
Systematic, stop-gradient (PyDPF)
Soft (α = 0.5)
OT (PyDPF)
OT (PFJAX)
OT (FilterFlow)
Eigen moment matching
Gaussian moment matching
Gumbel-softmax
Variational (KL)
(b) Paired difference from systematic on the same observation sequences
N = 32
table ±0.046
N = 128
table ±0.046
N = 512
table ±0.048
N = 2048
table ±0.048
reference
-0.02 0 0.02
-0.01 0 0.01
Paired difference from systematic (+ = worse than systematic)
(c) Where the state MSE comes from
N = 32
N = 128
7%
7%
8%
9%
7%
7%
7%
7%
8%
6%
6%
6%
6%
7%
7%
7%
2%
2%
4%
4%
2%
2%
2%
2%
2%
2%
2%
2%
2%
2%
2%
2%
0.00 0.05 0.10 0.15 0.20 0.25 0.30
Mean squared error: gray = exact Kalman filter, colour = added by the particle filter